Practical - 7

Flow Control at Data Link Layer

Aim :-

Write a program to implement flow control at data link Layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another

Create a sender program with following features :-

1. Input Window size from the user.
2. Input a Text message from the user
3. Consider 1 character per frame.
4. Create a frame with the following fields [Frame no., DATA].
5. Send the frames
6. Wait for the acknowledgement from the Receiver
7. Reader a file called Receiver_buffer
8. Check ACK field for the Acknowledgement number.
9. If the acknowledgement number is as expected, send new set of frames accordingly. Else if NACK is received, resend the frames accordingly.

Receiver program

1. Reader a file called Sender_Buffer.
2. Check the frame no.
3. If the Fame no. are as expected, write the appropriate ACK no. in the Receiver_Buffer File. Else write NACK no in the Receiver_Buffer file

Code

Sender.py

```
Import time
import os
ws = int(input("Window size: "))
msg = input("Enter message: ")

Frames = [(i+1, ch) for i, ch in enumerate(msg)]
i=0
while i < len(Frames):
    window = Frames[i : i + ws]

    with open("Sender_Buffer", "w") as f:
            for no, data in window:
                f.write(f"{no},{data}\n")

    print("\n Sending:", window)
    time.sleep(2)
    while not os.path.exists("Receiver_Buffer", "r") as f:
        ~~ack, no = f.read().strip().split(",")~~
        ~~no =~~
        time.sleep(1)

    with open("Receiver_Buffer", "r") as f:
        ack, no = f.read().strip().split(",")
        no = int(no)
        print("Received:", ack, no)
        if ack == "ACK" and no == window[-1][0]:
            i += ws
```

```
        elif ack == "NACK":
            i = no - 1

        os.remove("Receiver_Buffer")
print("\n Transmission completed.")
```

receiver.py

```
import time
import os
expected = 1
while True:
    if not os.path.exists("Sender_Buffer"):
        time.sleep(1)
        continue
    with open("Sender_Buffer", "r") as f:
        frames = [[int(x.split(",")[0]),
    x.split(",")[1]]
        for x in f.read().strip().split("\n")]

    if not frames:
        time.sleep(1)
        continue
    print("\n Received:", frames)
```

```
for no, data in frame:
    if no != expected:
        with open("Receiver_Buffer", "w") as F:
            F.write(F"NACK, {expected}")
            break
    else:
        with open("Receiver_Buffer", "w") as F:
            F.write(F"ACK, {no}")
            expected += 1
            time.sleep(1)
    time.sleep(1)
```

Input

```
WINDOW size : 3
Enter Message : BRICKS
```

output - Sender Terminal

```
Window size : 3
Enter message : BRICKS
Sending : [(1,'B'),(2,'R'),(3,'I')]
Received : ACK 3
Sending : [(4,'K'),(5,'S')]
Received : ACK 5
Transmission completed
```

Output - Receiver Terminal

Received: [[1, 'B'], [2, 'R'], [3, 'I']]

Received: [[4, 'K'], [5, 'S']]

Result :-

Thus the code is implemented for sliding Window Protocol.